Adelaide Stroke Incidence Study

Declining Stroke Rates but Many Preventable Cardioembolic Strokes

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Background and Purpose—Stroke incidence rates are in flux worldwide because of evolving risk factor prevalence, risk factor control, and population aging. Adelaide Stroke Incidence Study was performed to determine the incidence of strokes and stroke subtypes in a relatively elderly population of 148 000 people in the Western suburbs of Adelaide.

Methods—All suspected strokes were identified and assessed in a 12-month period from 2009 to 2010. Standard definitions for stroke and stroke fatality were used. Ischemic stroke pathogenesis was classified by the Trial of ORG 10172 in Acute Stroke Treatment criteria.

Results—There were 318 stroke events recorded in 301 individuals; 238 (75%) were first-in-lifetime events. Crude incidence rates for first-ever strokes were 161 per 100 000 per year overall (95% confidence interval [CI], 141–183), 176 for men (95% CI, 147–201), and 146 for women (95% CI, 120–176). Adjusted to the world population rates were 76 overall (95% CI, 59–94), 91 for men (95% CI, 73–112), and 61 for women (95% CI, 47–78). The 28-day case fatality rate for first-ever stroke was 19% (95% CI, 14–24); the majority were ischemic (84% [95% CI, 78–88]). Intracerebral hemorrhage comprised 11% (8–16), subarachnoid hemorrhage 3% (1–6), and 3% (1–6) were undetermined. Of the 258 ischemic strokes, 42% (95% CI, 36–49) were of cardioembolic pathogenesis. Atrial fibrillation accounted for 36% of all ischemic strokes, of which 85% were inadequately anticoagulated.

Conclusions—Stroke incidence in Adelaide has not increased compared with previous Australian studies, despite the aging population. Cardioembolic strokes are becoming a higher proportion of all ischemic strokes. (***Stroke.* 2013;44:1226-1231.**)

Key Words: acute cerebral hemorrhage ■ acute cerebral infarction ■ acute stroke syndromes ■ epidemiology

One of the greatest healthcare challenges of the 21st century is the global aging population.[1] Although the incidence of stroke has been declining in the developed world during the past 30 years, population aging threatens to increase total stroke incidence and the burden of poststroke disability. One prediction, calculated from the age-specific stroke incidence reported in international studies, suggested that the worldwide total stroke burden would increase 5-fold from 13 million disability-adjusted life years in the 1990s to 68 million disability-adjusted life years by 2020.[2] However, age-specific stroke incidence rates may be declining sufficiently to compensate for changing demographics. Indeed, serial population stroke studies conducted during the past 3 decades have demonstrated decreasing stroke incidence despite increasing population age.[3] Increasing stroke burden is not an inevitable consequence for an aging population if primary prevention strategies are targeted to stroke incidence patterns.

The Adelaide Stroke Incidence Study (ASCEND) was funded by the National Health and Medical Research Council (Project grant 565402) as an adjuvant study to an ongoing contemporaneous (but of longer duration) rural stroke incidence study (Stroke Epidemiology in an Australian Rural Cohort). ASCEND was conducted in a postcode-defined region of the Western suburbs of Adelaide, South Australia (Figure 1), which is a region with a high proportion of elderly patients.

Methods

Study Population

The urban incidence data were collected between July 15, 2009, and July 15, 2010. According to projections from the Australian Bureau of statistics, this region had a total of 148 000 residents during the study period. In our population, 19% of residents were >65, 10% >75, and 3% were >85 years of age. This is significantly higher than the rest of Adelaide (15%, 8%, and 2%, respectively) and also higher than

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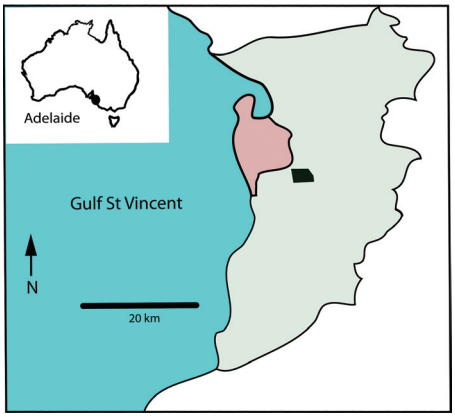

Figure 1. Map of Adelaide Statistical Division (shaded) and study region (red). The Adelaide central business district is shaded black.

the most recent Australian stroke incidence study (16%; >65 years of age). Aboriginal or Torres Strait Islanders accounted for 5% of the population. Income in the study region was one-third less than the average for Adelaide as a whole, with similar rates of unemployment. Most residents in the study region were born in Australia (74%), with 18% born in Europe and 6% born in Asia, which is similar to the rest of Adelaide. Only 5% of residents had migrated to Australia in the past 15 years.

Definition

Stroke was defined by the World Health Organization (WHO) as rapidly developing clinical signs of focal (or global) disturbance of cerebral function lasting >24 hours (unless interrupted by surgery or death) with no apparent cause other than of vascular origin.[4] All cases of nontraumatic subarachnoid hemorrhages (SAHs) were included. The definition excluded brain tumor, subdural hematoma, trauma, postseizure palsy, and transient ischemic attack. Cases of transient symptoms with infarction were recorded during ascertainment but not included in reported stroke incidence to be consistent with the WHO definition of stroke. Traumatic hemorrhages were also recorded but not included in reported stroke incidence to be consistent with the WHO definition.

Ascertainment

A stroke neurologist (J.L.) and 2 study nurses (S.C. and J.C.) participated in case ascertainment after thorough training in data collection and ascertainment procedure. Every emergency department presentation or hospital admission in any hospital in the city of Adelaide was screened for potential strokes. Every nursing home in the study region was contacted weekly, and any potential case of stroke or transient ischemic attack was investigated by patient visit. The stroke outpatient clinic in the region was monitored, and neurology outpatient clinics in the region were involved in the study. General practitioners were contacted every 3 months and posters displayed. General physicians were contacted and seminars were presented at all tertiary referral hospitals. All public hospital radiology referrals and results were screened, and discharge coding was screened in all hospitals. The death register was examined and any possible cases investigated. Any cases referred for possible stroke within the region were investigated.

Case Assessment

A standardized case history and examination (The National Institutes of Health Stroke Scale [NIHSS]) was performed. Each case history included an extensive medical history, the history of the stroke presentation, and collation of all investigations and medical records. When the patient could not be interviewed, the next-of-kin was contacted, and medical records were reviewed.

Atrial fibrillation (AF) was classified as known AF for which an ECG-confirmed diagnosis of AF or paroxysmal AF (PAF) had been made at any time before the stroke. New AF or new PAF was defined as ECG or cardiac telemetry–confirmed AF or PAF that was identified after the stroke.

Each case was assessed against inclusion criteria by 2 neurologists, with reference to a third neurologist in the case of disagreement. Cases were included in which a diagnosis of probable stroke could be made by consensus. Possible strokes were excluded. Ischemic stroke cases were classified according to the Trial of ORG 10172 in Acute Stroke Treatment (TOAST) criteria[5] (please see online-only Data Supplement). Patients were classified as recurrent stroke if there was history of previous stroke. This also included cases who had >1 stroke presentation within the study period. All other cases were classified as first-ever-in-a-lifetime stroke.

Data Collection and Calculation of Rates

All investigations, clinical details, and results were recorded onto a purpose-built database. All brain and vascular imaging was obtained and reviewed by the authors. Incidence rates (first-ever) and attack rates (all strokes) were calculated per 100 000 per year with confidence intervals (CIs) calculated from the Poisson distribution. Rates were standardized to the world population to allow interstudy comparisons.[6] Case fatality rates report death within 28 days of stroke. Data are reported with 95% CI.

Ethics Approval

This study obtained ethical approval from every private and public hospital in Adelaide and from the University of Adelaide. Informed consent was obtained from all patients or the next-of-kin when the patient was unable to consent. No patients (or proxies) refused to participate.

Results

Ascertainment

A total of 312 patients were initially recruited to the study. Eleven patients were judged to be possible rather than probable strokes after case review by the diagnostic panel and were excluded.

Comprehensive surveillance from all sources (Figure 2; ascertainment sources) led to a total of 1397 patients being seen, of whom 1096 were excluded (Table 1). Of these, 191 patients had transient symptoms lasting <24 hours (including 1 thrombolysis case), 905 were not attributable to stroke, and 19 cases had radiological evidence of cerebral infarction but did not meet the WHO stroke definition. TSI accounted for 19 exclusions.

Of the 301 patients with a stroke occurring within the study period, 281 (93%) were ascertained by >1 independent source, and 240 (80%) by >2. Hospital presentation for the stroke occurred in 289 patients (96%). Despite thorough out-of-hospital case ascertainment, few strokes were managed in the community, suggesting a change in Australian practice. Of all recruited cases, 73% were examined within 1 week of symptom onset, which was substantially more than previous Australian studies.[7] Retrospective data were collected in 28 events in which no examination was possible. The median NIHSS was 6 (interquartile range, 2–14). Neuroimaging was performed in 301 strokes (95% of total), including 229 (96%) of patients with first-ever stroke. No imaging was performed after 17 strokes, computed tomography only in 188, MRI only in 15, and in 98 events, both computed tomography and MRI were obtained. Two hundred sixty-six strokes were confirmed with any neuroimaging. In 10 events, stroke was diagnosed despite negative MRI. In 25 strokes, the diagnosis was made despite negative computed tomography scan. Imaging of neck vessels was performed in 86% of all ischemic strokes.

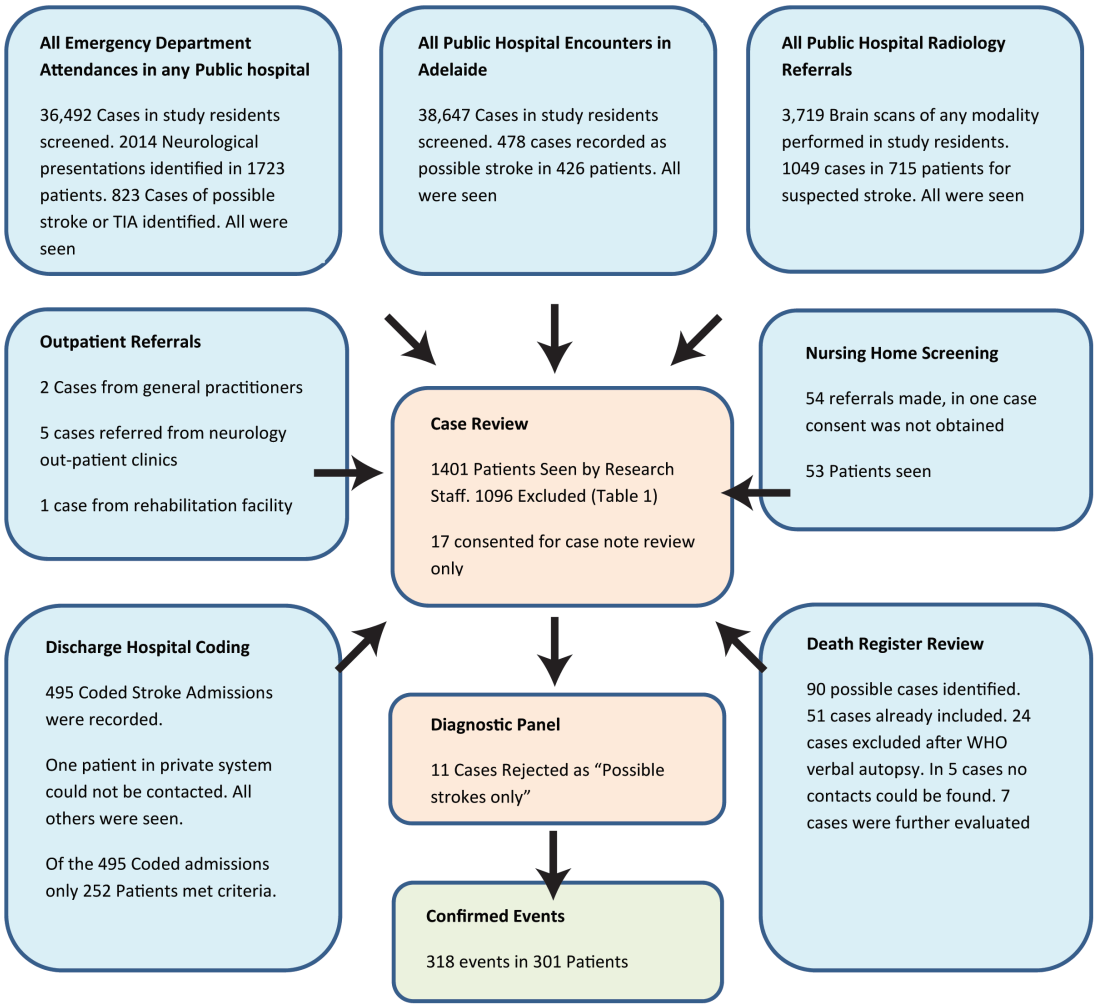

Figure 2. Ascertainment sources.

Incidence

Of 301 individuals, 318 strokes were recorded, giving a rate of 215 per 100 000 person years. Incidence by age per 100 000 person years varied from 60 (95% CI, 47–75) <65 years of age to 2251 (1823–2749) >85 years of age (Table 2). Stroke attack rates were nonsignificantly higher in men in most age groups (please see online-only Data Supplement).

Of these, 238 (74.8%) were first-ever-in-a-lifetime events (Table 3). Crude incidence rates per 100 000 person years for first-ever strokes were 161 overall (95% CI, 141–183), 176 for men (95% CI, 147–201), and 146 for women (95% CI, 120–176). Adjusted to the world population, incidence rates per 100 000 person years were 76 overall (95% CI, 59–94), 91 for men (95% CI, 73–112), and 61 for women (95% CI, 47–78).

Stroke Risk Factors

Hypertension had been diagnosed in 75% of cases before the stroke with 58% on ≥1 antihypertensive medication. Hypercholesterolemia had been diagnosed in 41% of cases before the stroke, with 36% taking lipid-lowering medication. A history of ischemic heart disease was present in 26% of cases. Diabetes mellitus had previously been diagnosed in 24% of cases. Current smokers accounted for 13% of cases. A previous transient ischemic attack was reported in 11% of cases. Body mass index could be determined in 86% of cases. Median body mass index was 26 (interquartile range, 22–30). In 48% of events, the patient was taking ≥1 antiplatelet agent. Further details of risk factors, medications, and definitions are published online (please see Online-only Data Supplement)

Stroke Subtypes

In 17 strokes, no imaging was performed and no subtype assigned. The majority of subtyped first-ever strokes were ischemic (84% [95% CI, 78–88]). Intracerebral hemorrhage comprised 11% (8–16), SAH 3% (1–6), and 3% (1–6) were undetermined.

Of the total 258 ischemic strokes, 16% (95% CI, 12%–22%) were from large artery atherosclerosis, 42% (95% CI, 36%–49%) were cardioembolic, 11% (95% CI, 7%–16%) attributable to small vessel disease, 6% (95% CI, 3%–10%) from other causes, and 25% (95% CI, 20%–32%) were undetermined (Table 2; Figure 3). First-ever and recurrent ischemic stroke subtypes are published online (please see online-only Data Supplement).

Case fatality at 28 days for all events was 19% (95% CI, 15%–24%). For first-ever events, case fatality was 18% (95% CI, 14%–24%), and for recurrent events, it was 21% (14%–32%).

Table 1. Referral Exclusions

Exclusion Reason	Number
Symptoms lasting <24 h	191
Outside study time or area	29
Excluded as symptoms not secondary to stroke	
Delirium or dementia	106
Acute infections	100
Psychiatric or psychogenic complaints	60
Syncope	64
Seizures	70
Dizziness, peripheral cause	53
Symptoms trauma related	47
Peripheral nerve disorder	37
Symptoms related to brain tumor	30
Symptoms from spinal cord compression	8
Symptoms secondary to migraine	33
Symptoms secondary to nonstroke movement disorder	5
Other systemic illness	263

Case fatality at 28 days for all ischemic strokes was 16%. Case fatality at 28 days for all hemorrhagic strokes was 35%. Case fatality by ischemic stroke subtype is published online (please see Online-only Data Supplement).

AF and Cardioembolic Stroke

A history of previous AF or PAF was identified in 78 stroke events. New onset AF was diagnosed at presentation in 26 events, and another 11 events were diagnosed with new onset PAF with cardiac monitoring.

Of 109 cardioembolic strokes, 81 were attributed to AF and 11 to PAF by the diagnostic panel. Of all AF-related strokes, 57 (70%) patients had been diagnosed before their event. Of these, 14 were therapeutically anticoagulated, 11 patients were subtherapeutically anticoagulated, and 32 patients were not anticoagulated. All 32 had a $CHADS_2$ score ≥2 before the event.[8] Of those 32, 16 had no contraindication to warfarin. Of the remaining 16, 2 had a history of gastrointestinal bleeding. For the remaining 14, treating doctors cited an unacceptably high risk of falling.

Warfarin and Intracranial Hemorrhage

Of the 43 nontraumatic hemorrhages observed, only 2 were taking warfarin at the time of the event and both were SAHs. Of 24 traumatic subdural hemorrhages, 3 patients were on warfarin. Five traumatic SAHs and 12 traumatic intracerebral hemorrhages occurred in the study population. No traumatic SAH or intracerebral hemorrhage patient was taking warfarin.

Discussion

ASCEND: A Novel Elderly Cohort

In our study, conducted between July 2009 and July 2010 in Western Adelaide, the crude incidence rate for first-ever stroke was 161 per 100 000 person years. One in 5 people in our population was >65 years of age. This population distribution matches projected trends for the United States and other developed countries during the next 20 years.[9] In the current study, 63% (95% CI, 58–68) of all strokes occurred in patients >75 years of age. In a 1996 Australian study, the proportion was 58% (53%–63%),[7] and in a 1988 Australian study, the proportion was 40% (32–48).[10] Our population was significantly older then recent populations studied in Dublin, Oxfordshire and only slightly older than Dijon.[11–13]

Strokes in patients who were >85 years of age accounted for 30% of first-ever strokes in our study compared with 16% in Dublin and 18% in Oxfordshire.[11,12] The proportion of first-ever ischemic stroke at ≥80 years of age was identical to the Dijon stroke study at 43%.[13]

Impact of Stroke Prevention Measures

Despite population aging, crude stroke incidence was not increased compared with previous stroke incidence studies

Table 2. Age- and Sex-Specific Stroke Attack Rates Per 100 000 Population Per Year in Adelaide Australia (2009–2010)

			Annual Attack Rates Per 100 000 Persons								
	Population at Risk		Men			Women			Both Sexes		
Age, y	Men	Women	n	Rate	95% CI	n	Rate	95% CI	n	Rate	95% CI
0–14	12 328	11 633	0	…	…	0	…	…	0	…	…
15–24	9193	8827	0	…	0–40	0	…	0–42	0	…	0–20
25–34	9225	9012	3	33	7–95	3	33	7–97	6	33	12–72
35–44	10 313	10 437	5	48	16–113	3	29	6–84	8	39	17–76
46–54	10 491	10 791	13	124	66–212	9	83	38–158	22	103	65–157
55–64	8908	9233	25	281	182–414	11	119	59–213	36	198	139–275
65–74	5996	6997	21	350	217–535	24	343	220–510	45	346	253–463
75–84	4516	5863	55	1218	917–1585	50	853	633–1124	105	1012	827–1225
≥85	1590	2675	45	2831	2065–3788	51	1907	1420–2507	96	2251	1823–2749
All ages	72 559	75 468	167	230	197–268	151	200	169–235	318	215	192–240
Standardized rate*				114	94–137		80	63–100		96	78–117

CI indicates confidence interval.

*Standardized to World Health Organization world population.[6]

Table 3. Age-Specific Incidence Rates (Per 100 000 Population Per Year) for First-Ever Lifetime Stroke by Subtype in Adelaide, Australia (2009–2010)

		Cerebral Infarction			ICH			SAH			Undetermined		
Age, y	Population at Risk	n	Rate	95% CI	n	Rate	95% CI	n	Rate	95% CI	n	Rate	95% CI
0–14	23 961	…	…	…	…	…	…	…	…	…	…	…	…
15–24	18 019	0	0	0–20	0	0	0–20	0	0	0–20	0	0	0–20
25–34	18 237	3	16	3–48	1	5	0–31	2	11	1–40	0	0	0–20
35–44	20 750	6	29	11–63	0	0	0–18	1	5	0–27	0	0	0–18
46–54	21 282	14	66	36–110	4	19	5–48	1	5	0–26	0	0	0–17
55–64	18 140	29	160	107–230	2	11	1–40	0	0	0–20	0	0	0–20
65–74	12 993	34	262	181–366	1	8	0–43	0	0	0–28	0	0	0–28
75–84	10 379	53	511	383–668	14	135	74–226	3	29	6–84	2	19	2–70
≥85	4264	60	1407	1074–1811	4	94	26–240	0	0	0–87	4	94	26–240
All ages	148 027	199	134	116–154	26	18	11–26	7	5	2–10	6	4	1–9
Standardized rate*			62	48–79		8	3–16		4	1–9		1	0–6

CI indicates confidence interval; ICH, intracerebral haemorrhages; and SAH, subarachnoid hemorrhage.

*Standardized to World Health Organization world population.[6]

within Australia. Crude stroke incidence per 100 000 person years in Perth was 191 in 1989, 157 in 1996, and 128 in 2001.[10] In Melbourne, crude stroke incidence per 100 000 person years was reported to be 206 in 1999.[7] In Adelaide, the crude stroke incidence per 100 000 person years was 161 in 2010, despite a much older population. Advances in primary prevention and population-based public health initiatives (such as smoking prevention) can substantially negate and even reverse the stroke epidemic predicted on crude demographic trends.[14] Nearly half of the events occurred in patients on antiplatelet agents. More than half were already taking antihypertensives. One third of strokes occurred in patients on statin therapy. Only 13% of stroke events occurred in current smokers.

First-ever stroke mortality in the current study (18%) trended lower than previous Australian studies (19% and 23%, respectively),[7,10] despite the advanced age and high cardioembolic stroke rates in our population (both strong predictors of stroke-related mortality).[14] Factors that may reduce case fatality include high rates of stroke unit care and prestroke pharmacotherapy.[14]

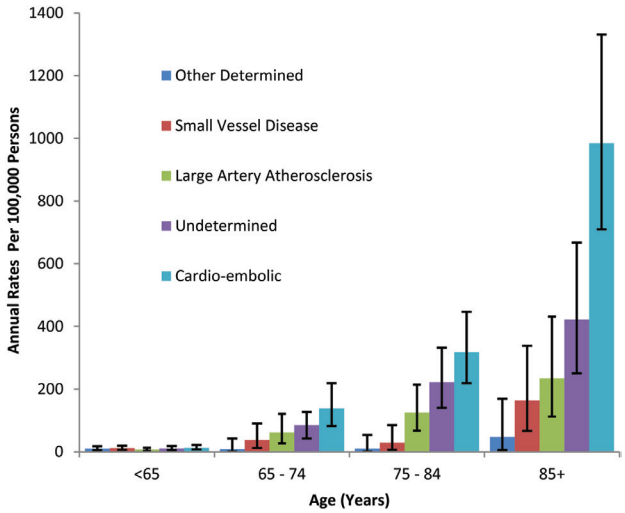

Figure 3. Age-specific incidence rates for all ischemic stroke subtypes in Adelaide (2009–2010).

Burden of Cardioembolic Stroke

Comparing the proportion of stroke attributable to cardioembolism between studies is complicated by variations in definition. The Dijon population stroke study of 2006 reported that 24% of events were cardioembolic using the TOAST definition.[13] However, the authors attributed 28% of all ischemic strokes to AF defined by a panel agreeing the likely cause independent of the TOAST criteria.[15] In the Dublin Stroke study, the presence of AF in association with a stroke, regardless of the cause, was reported in 31% of cases.[16] Applying the TOAST criteria to the Dublin study, 22% of all ischemic strokes were reported as cardioembolic from any cause.[11] In the Perth community stroke study of 1989, 15% of strokes were attributed to embolic infarction after review.[7] Despite variations in definition, it is apparent that the proportion of identified cardioembolic stroke by TOAST criteria (42%) is higher in Adelaide in 2010.

There are several potential explanations. First, our population was older than other studied populations (although only slightly higher than Dijon). Second, a reduction of lacunar and atherosclerotic stroke subtypes is likely from high statin use and low smoking rates in our population. Finally, our elderly population seems more intensively investigated than the elderly in previous studies.[17]

The $CHADS_2$ index is a widely used scheme to stratify cardioembolism risk and guide treatment.[8] If treatment guidelines had been followed, one third of cardioembolic strokes in our population may have been prevented. Although the most-cited warfarin contraindication was falls risk, in our population, no cases of intracerebral hemorrhage occurred in anticoagulated patients, and only 2 minor SAHs occurred. Our study provides population-based evidence that medical practitioners are overcautious in trying to avoid intracranial bleeding from anticoagulation. A comprehensive population-wide approach to AF, including newer fixed-dose agents, could substantially decrease stroke incidence in our population.

Limitations

Our study has several limitations. Although ascertainment was intensive, a small number of cases may have been missed. A small proportion of cases (9%) were identified retrospectively. In these patients, TOAST classification is more likely to conclude an undetermined cause. Our study was observational and therefore did not direct investigations. Where strokes were not investigated by imaging (5%), no classification was possible.

Summary

Stroke incidence in Adelaide has not increased compared with other Australian studies despite the aging population. In our study, cardioembolic stroke represented a high proportion of ischemic stroke. Many of these cardioembolic strokes were potentially preventable.

Sources of Funding

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Disclosures

None.

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